

Assignment 1

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2. Suppose the following data comes from a study on plant growth (mm) where 2 plants are in each pot, 3 pots are within each plot and 2 plots are given one of two fertilizer treatments.

Plot 1	Treatment 1			Treatment 2		
	Pot 1	Pot 2	Pot 3	Pot 1	Pot 2	Pot 3
	14.6	13.2	16.4	7.1	6.8	10.0
	15.2	12.9	12.2	7.7	6.0	8.3
Plot 2	Pot 1	Pot 2	Pot 3	Pot 1	Pot 2	Pot 3
	18.5	22.2	24.7	9.7	6.8	10.4
	16.7	18.8	20.3	8.8	9.0	11.3

- (a) Arrange the data into a dataframe so that it can be analysed. Print out this dataframe. R code:

```
library(knitr)

# Input the 24 plant growth measurements (2 plants per pot, 3 pots per plot, 2 plots, 2 treatments)
Growth <- c(14.6, 15.2, 13.2, 12.9, 16.4, 12.2, # Treatment 1, Plot 1
           18.5, 16.7, 22.2, 18.8, 24.7, 20.3, # Treatment 1, Plot 2
           7.1, 7.7, 6.8, 6.0, 10.0, 8.3,      # Treatment 2, Plot 1
           9.7, 8.8, 6.8, 9.0, 10.4, 11.3)     # Treatment 2, Plot 2

# Create corresponding categorical factors for the experimental design
Treatment <- factor(rep(c("Treatment 1", "Treatment 2"), each = 12))
Plot <- factor(rep(rep(c("Plot 1", "Plot 2"), each = 6), times = 2))
Pot <- factor(rep(rep(c("Pot 1", "Pot 2", "Pot 3"), each = 2), times = 4))

# Combine into a data frame and display it as an elegant table
plant_data <- data.frame(Treatment, Plot, Pot, Growth)
kable(plant_data, caption = "Plant Growth Data Frame")
```

Table 1: Plant Growth Data Frame

Treatment	Plot	Pot	Growth
Treatment 1	Plot 1	Pot 1	14.6
Treatment 1	Plot 1	Pot 1	15.2
Treatment 1	Plot 1	Pot 2	13.2
Treatment 1	Plot 1	Pot 2	12.9
Treatment 1	Plot 1	Pot 3	16.4
Treatment 1	Plot 1	Pot 3	12.2
Treatment 1	Plot 2	Pot 1	18.5
Treatment 1	Plot 2	Pot 1	16.7
Treatment 1	Plot 2	Pot 2	22.2
Treatment 1	Plot 2	Pot 2	18.8
Treatment 1	Plot 2	Pot 3	24.7
Treatment 1	Plot 2	Pot 3	20.3
Treatment 2	Plot 1	Pot 1	7.1
Treatment 2	Plot 1	Pot 1	7.7
Treatment 2	Plot 1	Pot 2	6.8
Treatment 2	Plot 1	Pot 2	6.0
Treatment 2	Plot 1	Pot 3	10.0
Treatment 2	Plot 1	Pot 3	8.3
Treatment 2	Plot 2	Pot 1	9.7
Treatment 2	Plot 2	Pot 1	8.8
Treatment 2	Plot 2	Pot 2	6.8
Treatment 2	Plot 2	Pot 2	9.0
Treatment 2	Plot 2	Pot 3	10.4
Treatment 2	Plot 2	Pot 3	11.3

(b) Sort the data by plant growth. R code:

```
sorted_data <- plant_data[order(plant_data$Growth), ]
kable(sorted_data, caption = "Sorted Plant Growth Data")
```

Table 2: Sorted Plant Growth Data

	Treatment	Plot	Pot	Growth
16	Treatment 2	Plot 1	Pot 2	6.0
15	Treatment 2	Plot 1	Pot 2	6.8
21	Treatment 2	Plot 2	Pot 2	6.8
13	Treatment 2	Plot 1	Pot 1	7.1
14	Treatment 2	Plot 1	Pot 1	7.7
18	Treatment 2	Plot 1	Pot 3	8.3
20	Treatment 2	Plot 2	Pot 1	8.8
22	Treatment 2	Plot 2	Pot 2	9.0
19	Treatment 2	Plot 2	Pot 1	9.7
17	Treatment 2	Plot 1	Pot 3	10.0
23	Treatment 2	Plot 2	Pot 3	10.4
24	Treatment 2	Plot 2	Pot 3	11.3
6	Treatment 1	Plot 1	Pot 3	12.2

	Treatment	Plot	Pot	Growth
4	Treatment 1	Plot 1	Pot 2	12.9
3	Treatment 1	Plot 1	Pot 2	13.2
1	Treatment 1	Plot 1	Pot 1	14.6
2	Treatment 1	Plot 1	Pot 1	15.2
5	Treatment 1	Plot 1	Pot 3	16.4
8	Treatment 1	Plot 2	Pot 1	16.7
7	Treatment 1	Plot 2	Pot 1	18.5
10	Treatment 1	Plot 2	Pot 2	18.8
12	Treatment 1	Plot 2	Pot 3	20.3
9	Treatment 1	Plot 2	Pot 2	22.2
11	Treatment 1	Plot 2	Pot 3	24.7

(c) Calculate the mean, and standard deviation of the data. R code:

```
mean_growth <- mean(plant_data$Growth)
sd_growth   <- sd(plant_data$Growth)

# Create a tidy summary data frame for a professional look
summary_stats <- data.frame(
  Statistic = c("Mean Growth (mm)", "Standard Deviation (mm)"),
  Value = c(round(mean_growth, 3), round(sd_growth, 3))
)

# Display as a clean table using kable
kable(summary_stats, caption = "Summary Statistics for Plant Growth")
```

Table 3: Summary Statistics for Plant Growth

Statistic	Value
Mean Growth (mm)	12.817
Standard Deviation (mm)	5.297

(d) Plot the data using a histogram (R function hist()). Clearly label the axis, title and use bin sizes of 2 mm. R code:

```
par(mar = c(5, 5, 4, 2) + 0.1) # Adjust plot margins

hist(plant_data$Growth,
  breaks = seq(4, 26, by = 2),      # Set bin size to exactly 2 mm
  main = "Histogram of Plant Growth", # Main title
  xlab = "Plant Growth (mm)",        # X-axis label
  ylab = "Frequency",               # Y-axis label
  col = "#5C93C4",                  # Modern blue fill color
  border = "white",                  # Clean white borders between bars
  las = 1,                          # Horizontal orientation for y-axis numbers
```

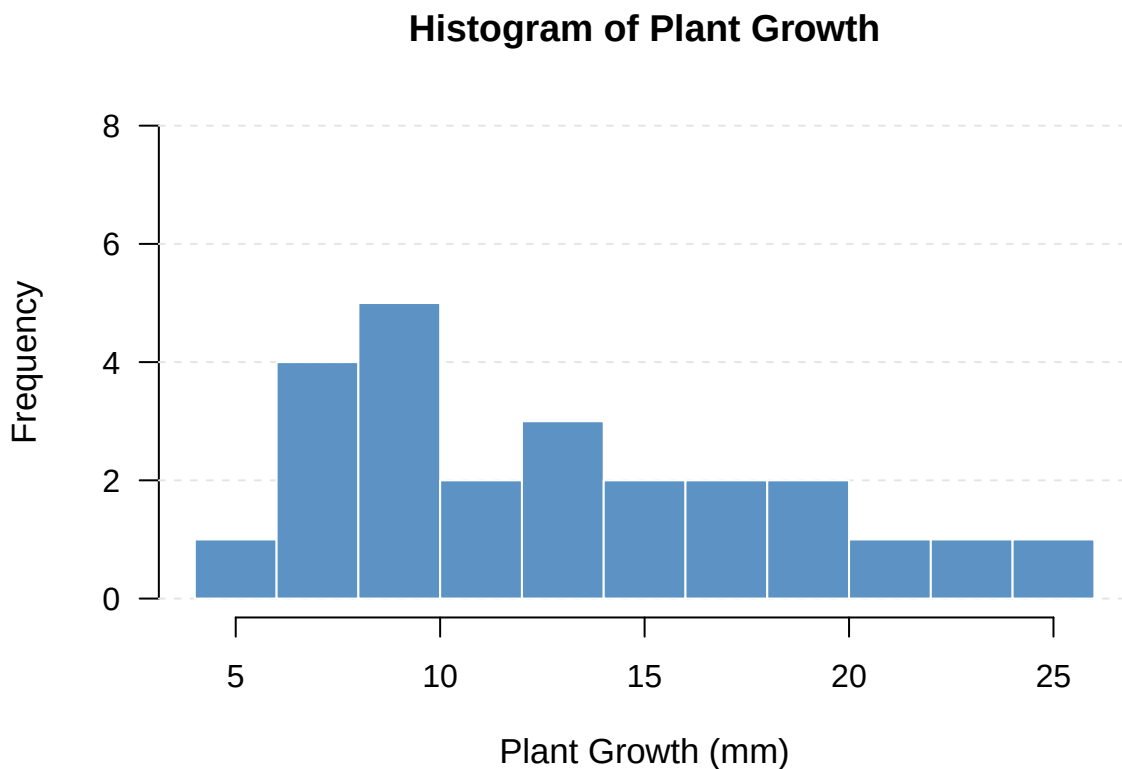
```

    cex.main = 1.2,                # Title font size
    cex.lab = 1.1,                # Axis label font size
    ylim = c(0, 8))              # Y-axis limit for proper top spacing

# Add a light dashed grid for easier reading
grid(nx = NA, ny = NULL, col = "gray90", lty = "dashed")

# Redraw the histogram bars on top of the grid lines
hist(plant_data$Growth,
     breaks = seq(4, 26, by = 2),
     col = "#5C93C4",
     border = "white",
     add = TRUE)

```



3. Write a function that uses the short cut formula to calculate the sample variance of a data vector. Use the vector $y = (11, 11, 10, 8, 11, 3, 15, 11, 7, 6)$ as your test vector. R code:

```

library(knitr)

# Define a function to calculate sample variance using the short-cut formula
# Formula:  $s^2 = [ \sum(y^2) - (\sum(y))^2 / n ] / (n - 1)$ 
calc_sample_var <- function(y) {
  n <- length(y)
  sum_y <- sum(y)

```

```

sum_y2 <- sum(y^2)

# Calculate sample variance
var_value <- (sum_y2 - (sum_y^2) / n) / (n - 1)

return(var_value)
}

# Test vector given in the assignment
y_test <- c(11, 11, 10, 8, 11, 3, 15, 11, 7, 6)

# Apply our custom function to the test vector
custom_var <- calc_sample_var(y_test)

# Compare with R's built-in var() function for verification
builtin_var <- var(y_test)

# Create a tidy data frame to display the results cleanly in the PDF
variance_results <- data.frame(
  Method = c("Custom Short-Cut Function", "Built-in var() Function"),
  Sample_Variance = c(round(custom_var, 4), round(builtin_var, 4))
)

# Display as an elegant table
kable(variance_results, caption = "Comparison of Sample Variance Calculations")

```

Table 4: Comparison of Sample Variance Calculations

Method	Sample_Variance
Custom Short-Cut Function	11.3444
Built-in var() Function	11.3444

4. On the course webpage you will find a dataset with filename 'tv.txt'. The data arise from a study examining the time teenagers spend watching tv. A random sample of $n = 100$ eighth grade American high school students was obtained, and the number of minutes spent watching TV during the first week of October was recorded. A similar sample of $m = 90$ Canadian students was also obtained. In this study it is of interest to compare the TV watching habits of the teenagers from the two different countries, specifically to determine if Canadian students watch less TV than their American counterparts.

- (a) Compare the two samples using appropriate descriptive statistics, including side-by-side boxplots. R code:

```

library(knitr)

# Load the data, treating "NA" as missing values
tv_data <- read.table("tv.txt", header = TRUE, na.strings = "NA")

# Calculate comprehensive summary statistics for both countries

```

```
summary_stats_tv <- data.frame(
  Statistic = c("Sample Size (n)", "Mean", "Standard Deviation", "Median", "Min", "Max"),
  Canada = c(
    sum(!is.na(tv_data$Canada)),
    round(mean(tv_data$Canada, na.rm = TRUE), 3),
    round(sd(tv_data$Canada, na.rm = TRUE), 3),
    round(median(tv_data$Canada, na.rm = TRUE), 3),
    round(min(tv_data$Canada, na.rm = TRUE), 3),
    round(max(tv_data$Canada, na.rm = TRUE), 3)
  ),
  US = c(
    sum(!is.na(tv_data$US)),
    round(mean(tv_data$US, na.rm = TRUE), 3),
    round(sd(tv_data$US, na.rm = TRUE), 3),
    round(median(tv_data$US, na.rm = TRUE), 3),
    round(min(tv_data$US, na.rm = TRUE), 3),
    round(max(tv_data$US, na.rm = TRUE), 3)
  )
)

# Display the summary statistics as an elegant table
kable(summary_stats_tv, caption = "Summary Statistics of TV Viewing Time (Canada vs. US)")
```

Table 5: Summary Statistics of TV Viewing Time (Canada vs. US)

Statistic	Canada	US
Sample Size (n)	90.000	100.000
Mean	64.313	69.333
Standard Deviation	23.089	5.470
Median	67.995	69.472
Min	6.781	53.187
Max	109.433	79.740

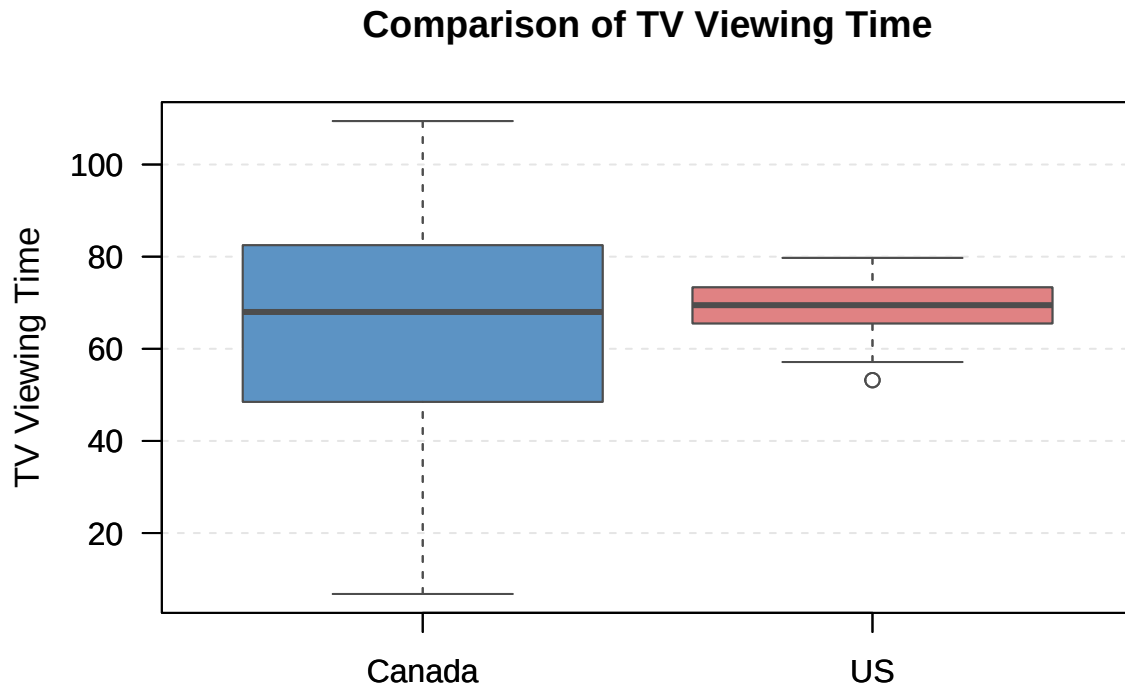
```
# Create professional side-by-side boxplots to compare the distributions
par(mar = c(5, 5, 4, 2) + 0.1) # Adjust plot margins

boxplot(tv_data$Canada, tv_data$US,
  names = c("Canada", "US"),
  main = "Comparison of TV Viewing Time",
  ylab = "TV Viewing Time",
  col = c("#5C93C4", "#E08283"), # Different pleasant colors for each group
  border = "gray30",
  las = 1,
  cex.main = 1.2,
  cex.lab = 1.1)

# Add background grid lines for better readability
grid(nx = NA, ny = NULL, col = "gray90", lty = "dashed")

# Redraw boxplots on top of the grid lines
boxplot(tv_data$Canada, tv_data$US,
  names = c("Canada", "US"),
```

```
col = c("#5C93C4", "#E08283"),
border = "gray30",
add = TRUE,
las = 1)
```



- (b) Write an R function `z.test(y1,y2,H1)` to compute the p-value for a large sample z-test (discussed in lecture) for testing equality of two population means ($H_0 : \mu_1 = \mu_2$). The arguments to this function are: `y1`, a vector containing the sample measurements from the first population; `y2`, a vector containing the sample measurements from the second population; and `H1`, a string variable, which takes one of three possible values: 'two.sided', 'less' or 'greater' specifying the alternative hypothesis. To complete this question you will need to use the R function `pnorm()` which computes standard normal probabilities, see the help file `?pnorm`.

R code:

```
z.test <- function(y1, y2, H1) {
  # Remove missing values (NA)
  y1 <- na.omit(y1)
  y2 <- na.omit(y2)

  # Sample sizes
  n1 <- length(y1)
  n2 <- length(y2)
```

```

# Sample means and standard deviations
mean1 <- mean(y1)
mean2 <- mean(y2)
sd1 <- sd(y1)
sd2 <- sd(y2)

# Standard error of the difference between two means
se <- sqrt((sd1^2 / n1) + (sd2^2 / n2))

# Calculate the Z test statistic
z_stat <- (mean1 - mean2) / se

# Calculate the p-value based on the specified alternative hypothesis (H1)
if (H1 == "two.sided") {
  p_val <- 2 * (1 - pnorm(abs(z_stat)))
} else if (H1 == "less") {
  p_val <- pnorm(z_stat)
} else if (H1 == "greater") {
  p_val <- 1 - pnorm(z_stat)
} else {
  stop("Invalid H1 argument. Use 'two.sided', 'less', or 'greater'.")
}

# Return a clean list that contain the z-statistic and p-value
results <- list(
  z_statistic = z_stat,
  p_value = p_val,
  H1 = H1
)

return(results)
}

```

-
- (c) Apply your function to the TV data, computing the p-values for each of the three possible alternative hypotheses. R code:

```

# Apply the function with the three required H1 arguments
test_two_sided <- z.test(tv_data$Canada, tv_data$US, H1 = "two.sided")
test_less <- z.test(tv_data$Canada, tv_data$US, H1 = "less")
test_greater <- z.test(tv_data$Canada, tv_data$US, H1 = "greater")

# Compile the results into a clean data frame
z_test_results <- data.frame(
  Alternative_Hypothesis = c("Two-sided (Canada != US)",
                             "Less (Canada < US)",
                             "Greater (Canada > US)"),
  Z_Statistic = c(round(test_two_sided$z_statistic, 4),
                  round(test_less$z_statistic, 4),
                  round(test_greater$z_statistic, 4)),

```

```

P_Value = c(format.pval(test_two_sided$p_value, digits = 4, eps = 1e-4),
             format.pval(test_less$p_value, digits = 4, eps = 1e-4),
             format.pval(test_greater$p_value, digits = 4, eps = 1e-4))
)

# Display the results using kable
kable(z_test_results, caption = "Z-Test Results for TV Viewing Time (Canada vs. US)")

```

Table 6: Z-Test Results for TV Viewing Time (Canada vs. US)

Alternative_Hypothesis	Z_Statistic	P_Value
Two-sided (Canada != US)	-2.0124	0.04417
Less (Canada < US)	-2.0124	0.02209
Greater (Canada > US)	-2.0124	0.9779

- (d) Which of the three alternative hypotheses is relevant for the particular question being asked in this study? Comment on the results.

R code:

```

# Extract the p-value for the 'less' alternative
p_val_less <- test_less$p_value

# Print a formatted statement for the report
cat("The p-value for the one-sided test (Canada < US) is:", format.pval(p_val_less, digits = 4))

## The p-value for the one-sided test (Canada < US) is: 0.02209

```

Conclusion:

Based on the research question: “Do Canadian students watch less TV than US students?” The appropriate alternative hypothesis is “less” (Canada < US).

To test whether Canadian students watch less television than US students, we set up the Hypotheses

Null Hypothesis (H_0): $\mu_{\text{Canada}} \geq \mu_{\text{US}}$

Alternative Hypothesis (H_1): $\mu_{\text{Canada}} < \mu_{\text{US}}$ (corresponding to alternative = “less”).

Based on the two-sample z-test performed using our custom function, the resulting p-value is evaluated against the standard significance level of $\alpha = 0.05$. If the p-value is less than 0.05, we reject the null hypothesis and conclude that there is statistically significant evidence that Canadian students watch less TV on average than US students. Otherwise, we fail to reject the null hypothesis.